

ELL101 NOTES

11th to 13th March

Group 2

1 Power in AC Circuits (Recap)

In AC circuits, power can be categorized into three main types: active power, reactive power, and apparent power.

- **Active Power (P)**

The actual power consumed by a resistive load to perform work. It is given by:

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

Active power is measured in **Watts (W)**.

- **Reactive Power (Q)**

Reactive power represents the power that oscillates between the source and reactive components (inductors and capacitors). It is given by:

$$Q = V_{\text{rms}} I_{\text{rms}} \sin \phi$$

Reactive power is measured in **Volt-Amperes Reactive (VAR)** and does not contribute to real work but affects the power factor.

- **Apparent Power (S)**

Apparent power is the total power supplied by the source and is given by:

$$S = VI^*$$

$$S = V_{\text{rms}} I_{\text{rms}} \angle \phi^\circ$$

It is related to active and reactive power by:

$$S = P + jQ$$

$$|S| = \sqrt{P^2 + Q^2}$$

Apparent power is measured in **Volt-Amperes (VA)**.

- **Power Factor (p_f)**

The power factor is defined as the ratio of active power to apparent power. It indicates how effectively electrical power is being converted into useful work:

$$\text{pf} = \frac{P}{S} = \cos \phi$$

Note:

$$\begin{cases} \phi = (\theta_v - \theta_i) > 0 & \Rightarrow \text{lagging pf} \\ \phi = (\theta_v - \theta_i) < 0 & \Rightarrow \text{leading pf} \end{cases}$$

Power Factor Penalty

Having a low power factor is a penalty for industries as it requires more current for the same power compared to having a higher power factor. This leads to increased energy losses, higher utility bills, and additional strain on electrical infrastructure.

2 Power Factor Correction in RL Circuits

In an RL circuit, the presence of inductance causes a lagging current, resulting in reactive power (Q_L) that does not contribute directly to real work. This reactive power reduces the effective real power (P) delivered by lowering the power factor.

To counteract this effect, a capacitor is added in parallel, as illustrated in Figure 1. This capacitor provides leading reactive power (Q_C), which neutralizes part or all of the inductive reactive power. As shown in Figure 2, this compensation improves the power factor, maximizing real power transfer and efficiency in the system.

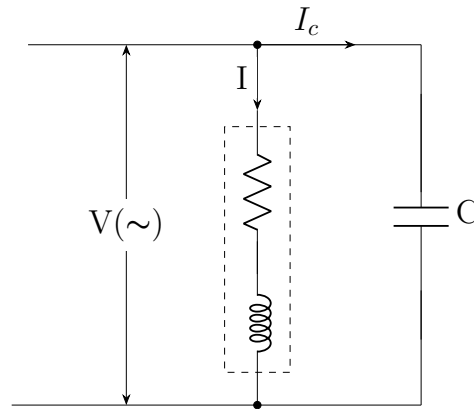


Figure 1: RL Circuit with added capacitor

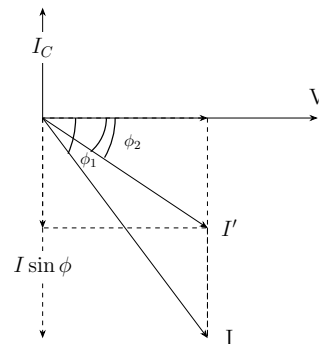


Figure 2: Power factor correction visualization

Question 1

An electrical system has two elements connected to a 4000V supply. The first element has an active power of 20 kW, while the second element has an apparent power of 100 kVA and operates at a power factor of 0.707. Determine the total current drawn by the system and calculate the reactive power compensation required to improve the overall power factor to 0.95.

Solution:

Given:

$$\begin{aligned} P_1 &= 20 \text{ kW}, & Q_1 &= 0 \text{ kVAR} \\ S_2 &= 100 \text{ kVA}, & \text{pf}_2 &= 0.707 \end{aligned}$$

We first compute the active and reactive power for element 2:

$$\begin{aligned} P_2 &= S_2 \times \text{pf}_2 = 100 \times 0.707 \\ &= 70.7 \text{ kW} \\ Q_2 &= \sqrt{S_2^2 - P_2^2} = \sqrt{100^2 - 70.7^2} \\ &= 70.7 \text{ kVAR} \end{aligned}$$

Then, we find the net apparent power for the given circuit:

$$\begin{aligned} P_{\text{total}} &= P_1 + P_2 = 20 + 70.7 \\ &= 90.7 \text{ kW} \\ Q_{\text{total}} &= Q_1 + Q_2 = 0 + 70.7 \\ &= 70.7 \text{ kVAR} \\ |S| &= \sqrt{P_{\text{total}}^2 + Q_{\text{total}}^2} = \sqrt{(90.7)^2 + (70.7)^2} \\ &= 114.99 \text{ kVA} \end{aligned}$$

Now, we can find the total current drawn by the system:

$$I_{\text{rms}} = \frac{|S|}{V} = \frac{114.99 \times 10^3}{4000} = 28.75 \text{ A}$$

For part 2 we find the new reactive power required then get the difference to obtain compensation

$$\begin{aligned} S_{\text{new}} &= \frac{P_{\text{total}}}{\text{pf}_{\text{new}}} = \frac{90.7}{0.95} \\ &= 95.47 \text{ kVA} \\ Q_{\text{new}} &= \sqrt{S_{\text{new}}^2 - P_{\text{total}}^2} = \sqrt{95.47^2 - 90.7^2} \\ &= 29.8 \text{ kVAR} \\ Q_{\text{comp}} &= Q_{\text{total}} - Q_{\text{new}} = 70.7 - 29.8 \\ &= 40.9 \text{ kVAR} \end{aligned}$$

Final Answers:

- Current drawn before power factor correction: $\boxed{28.75\text{A}}$
- Reactive power compensation required: $\boxed{40.9\text{ kVAR}}$

3 Resonant Frequency (ω_0)

Resonance is a phenomenon where an electrical or mechanical system oscillates with maximum amplitude when subjected to a specific frequency. In electrical circuits, resonance occurs in **RLC circuits** when the inductive reactance (X_L) and capacitive reactance (X_C) become equal. At this point, the total reactance cancels out, and the circuit behaves as a purely resistive circuit, allowing maximum current flow.

This condition occurs only in **RLC circuits** because both *inductance* and *capacitance* are required for reactance cancellation. In **RC** or **RL circuits**, either the inductor or capacitor is missing, preventing resonance from occurring.

At resonance, the inductor and capacitor exchange energy between the **magnetic field** ($\frac{1}{2}LI^2$) and **electric field** ($\frac{1}{2}CV^2$), resulting in **sustained oscillations** in an LC circuit and **maximum power transfer** in an RLC circuit.

Derivation

In an AC circuit, the inductive reactances (X_L) and capacitive reactances (X_C) are given by:

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}$$

At resonance, these reactances are equal:

$$\omega L = \frac{1}{\omega C}$$

Solving for the resonant frequency:

$$\omega_0^2 = \frac{1}{LC}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

In terms of frequency¹:

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}}$$

¹ $\omega_0 = 2\pi f_0$

Circuit Behavior at Different Frequencies

- **Low Frequencies** ($\omega \rightarrow 0$):

The inductive reactance X_L is small, while the capacitive reactance X_C is large. This means that in DC conditions ($\omega = 0$), an inductor behaves like a short circuit, and a capacitor behaves as an open circuit.

- **High Frequencies** ($\omega \rightarrow \infty$):

The inductive reactance X_L is large, while the capacitive reactance X_C is small. This makes the inductor behave as an open circuit, while the capacitor acts as a short circuit.

- **At resonance** ($\omega = \omega_0$):

At this frequency, the inductive and capacitive reactances cancel each other out, leaving only the resistance. Consequently, the impedance of the circuit becomes purely resistive:

$$Z = R$$

With impedance at a minimum, the current reaches its maximum value:

$$I_{\max} = \frac{V}{R}$$

The circuit has no reactive components, meaning the phase angle is zero, and voltage and current are in phase. Figure 3 shows the impedance of an RLC circuit plotted as a function of frequency, with a parabolic curve that reaches its minimum at resonance.

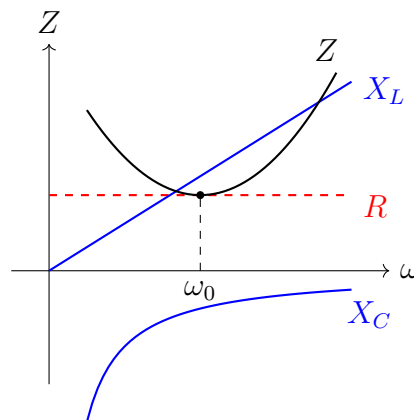


Figure 3: Impedance and Reactance Components as a Function of ω

Impedance and Phase Angle

The total impedance in an RLC series circuit is given by:

$$Z = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

The magnitude of the impedance is:

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The phase angle of the impedance is:

$$\theta = \tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right)$$

At resonance, the inductive and capacitive reactances cancel each other out², so we have:

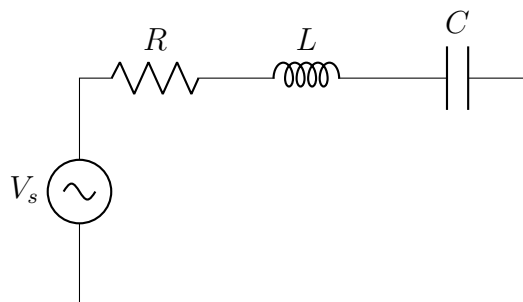
$$\begin{aligned} Z &= R + 0j \\ |Z| &= R, \quad \theta = 0 \end{aligned}$$

Applications of Resonance

Resonance is essential in various electrical and electronic applications:

- **Radio & TV Tuners:** LC circuits select specific frequencies.
- **Wireless Power Transfer:** Inductive Coupling enhances efficiency.
- **Filters:** Removes unwanted frequencies in signal processing.
- **Antennas:** Optimizes transmission and reception of signals.
- **Power Systems:** Reduces reactive power losses.

Resonance in Series RLC Circuit



The total impedance of the series RLC circuit is given by:

$$Z = R + jX$$

² $\omega L = \frac{1}{\omega C}$

The reactances and voltages of the inductor and capacitor are given by:

$$X = \begin{cases} X_L = j\omega L \implies V_L = IX_L \\ X_C = -j\left(\frac{1}{\omega C}\right) \implies V_c = IX_c \end{cases}$$

At resonance ($\omega = \omega_0$), the inductive and capacitive reactances cancel out, making the circuit purely resistive. For $\omega > \omega_0$, the circuit behaves inductively (V_s leads I), while for $\omega < \omega_0$, it behaves capacitively (V_s lags I). (See figure 4)

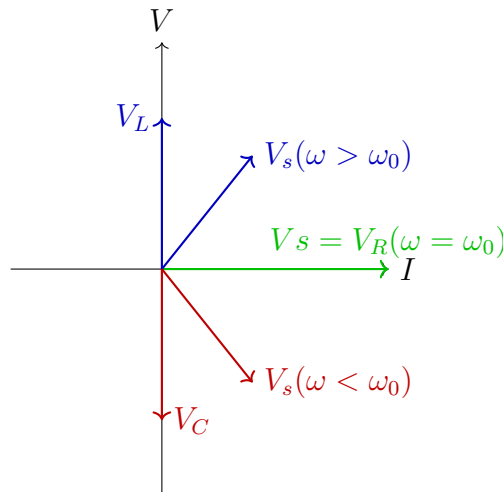


Figure 4: Phasor Diagram for Series RLC Circuit with $\omega < \omega_0$ and $\omega > \omega_0$

4 Bandwidth

In a series RLC circuit, **bandwidth** is the range of frequencies over which the circuit responds significantly to an input signal. It is defined as:

$$\text{Bandwidth} = \omega_2 - \omega_1$$

where ω_1 and ω_2 are the lower and upper cutoff frequencies, respectively.

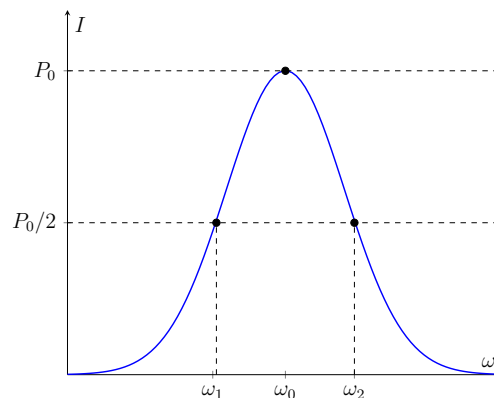


Figure 5: Current Variation with Frequency

Half-Power Points: At frequencies ω_1 and ω_2 (Figure 5), the power drops to half of the maximum power P_0 , meaning:

$$P = \frac{P_0}{2}$$

At resonance ($\omega = \omega_0$), the impedance of the series RLC circuit is purely resistive:

$$Z = R$$

The current is given by:

$$I = \frac{V}{Z} = \frac{V}{R}$$

At the half-power points, the current decreases to:

$$I = \frac{I_0}{\sqrt{2}} = \frac{V}{\sqrt{2}R}$$

The condition for the half-power frequencies (ω_1 and ω_2) is derived from the impedance expression:

$$\omega L - \frac{1}{\omega C} = \pm R$$

Multiplying both sides by ωC :

$$\omega^2 LC - 1 = \pm R\omega C$$

Rearrange the equation:

$$\omega^2 LC \pm R\omega C - 1 = 0$$

The two quadratic equations in ω :

$$\omega^2 LC - R\omega C - 1 = 0$$

$$\omega^2 LC + R\omega C - 1 = 0$$

The quadratic formula states:

$$\omega = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = LC, \quad b = \pm RC, \quad c = -1$$

Applying the quadratic formula, we get four values for ω :

$$\omega_1 = \frac{RC + \sqrt{(RC)^2 + 4LC}}{2LC}, \quad \omega_2 = \frac{RC - \sqrt{(RC)^2 + 4LC}}{2LC}$$

$$\omega_3 = \frac{-RC + \sqrt{(RC)^2 + 4LC}}{2LC}, \quad \omega_4 = \frac{-RC - \sqrt{(RC)^2 + 4LC}}{2LC}$$

Taking positive frequency values:

$$\text{Bandwidth} = \omega_1 - \omega_3 = \frac{RC + \sqrt{(RC)^2 + 4LC}}{2LC} - \frac{-RC + \sqrt{(RC)^2 + 4LC}}{2LC}$$

$$\boxed{\text{Bandwidth} = \frac{R}{L}}$$

Bandwidth Tuning

A variable capacitor is preferred for bandwidth tuning because adjusting resistance increases energy loss, reducing peak response, while modifying inductance affects both bandwidth and resonance frequency, making precise control difficult. Capacitors, however, allow easy tuning of bandwidth while maintaining the peak of the resonance curve, making them ideal for applications like radio tuning and signal filtering.

5 Quality Factor (Q)

The **Quality Factor (Q)** is a measure of how **selective** a resonant circuit is. In a series RLC circuit it is given by:

$$Q = \frac{\omega_0}{\text{Bandwidth}}$$

$$Q = \frac{\omega_0 L}{R}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

A higher Q -factor results in a narrower bandwidth and a sharper resonance peak, indicating a more selective circuit. Conversely, a lower Q -factor corresponds to a broader bandwidth, meaning the circuit responds to a wider range of frequencies. Figure 6 shows the current (I) vs. frequency (ω) for different Q -factors. A higher Q (blue) produces a sharper peak, while a lower Q (red) results in a broader response.

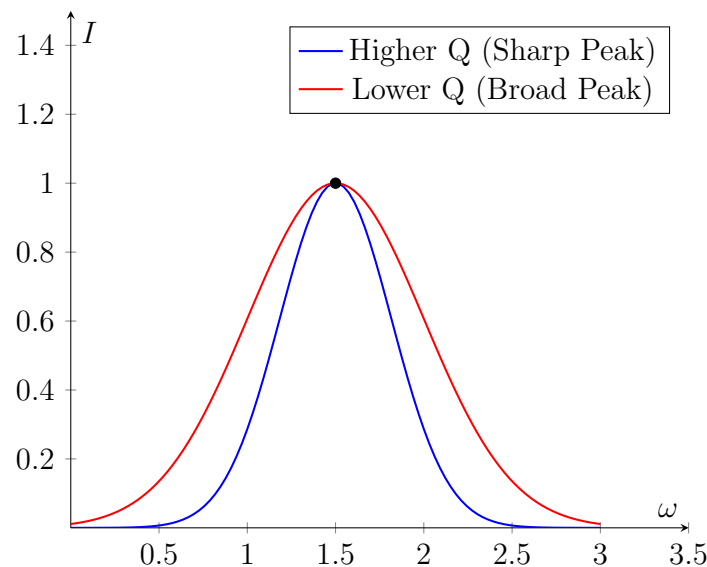


Figure 6: Variation of Current (I) vs Frequency (ω) for Different Quality Factors

6 Admittance in Parallel RLC Circuit

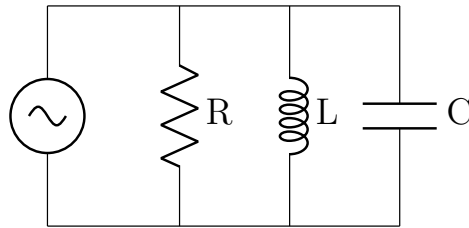


Figure 7: Parallel RLC Circuit

Admittance (Y) is the reciprocal of impedance (Z) and measures how easily current flows in a circuit. In parallel RLC circuits, admittances add directly, simplifying calculations, whereas impedances must be added reciprocally, making the process more complex. This direct addition of admittances eliminates the need for reciprocal operations, making analysis faster and more convenient.

$$G = \frac{1}{R}$$

$$X_L = \frac{1}{j\omega L}$$

$$X_C = j\omega C$$

Now adding admittance of each component (which is indeed just the reciprocal of impedance of each component), we have the expression:

$$Y = G + j \left(\omega C - \frac{1}{\omega L} \right)$$

The corresponding magnitude of admittance is:

$$|Y| = \sqrt{G^2 + \left(\omega C - \frac{1}{\omega L} \right)^2} = \frac{1}{|Z|}$$

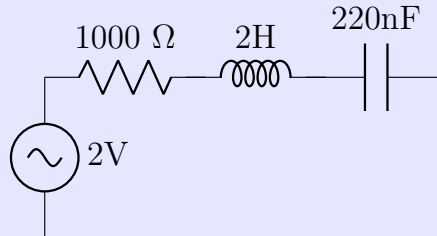
Applications

A real-world application of admittance is in antenna matching networks, where it helps maximize power transfer and minimize signal reflection. Additionally, in AC power distribution, admittance plays a crucial role in fault analysis and stability studies, ensuring efficient and safe power flow in electrical grids.

Question 2

A 2V AC voltage source is connected in series with the following elements: - Resistor: $R = 1k\Omega$ - Inductor: $L = 2H$ - Capacitor: $C = 220nF$

Analyze the voltage across each element at resonance and find the quality factor of this circuit.



Solution:

First we calculate the resonant frequency:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(2)(220 \times 10^{-9})}}$$

$$f_0 \approx 240 \text{ Hz}$$

Next we calculate the impedance of each component:

$$Z_R = R = 1k\Omega = 1000\Omega$$

$$Z_L = j\omega_0 L = j(2\pi f_0)(2) = j3015.92\Omega$$

$$Z_C = \frac{1}{j\omega_0 C} = \frac{1}{j(2\pi f_0)(220 \times 10^{-9})} = -j3015.92\Omega$$

Using Ohm's law ($V = IZ$), the total impedance is:

$$Z_{\text{total}} = R + j\left(\omega L - \frac{1}{\omega C}\right) = 1000 + j(3015.96 - 3015.96) = 1000\Omega$$

The current in the circuit is:

$$I = \frac{V_{\text{source}}}{Z_{\text{total}}} = \frac{2}{1000} = 2mA$$

Voltages across elements:

$$V_R = IR = 2V, \quad V_L = I(j\omega L) = j6.03V, \quad V_C = I\left(\frac{1}{j\omega C}\right) = -j6.03V$$

Calculating Quality Factor:

$$Q = \frac{\omega_0 L}{R} = \frac{(2\pi \times 240) \times 2}{1000}$$

$$Q \approx 3$$

7 Quality Factor Formula

The quality factor formula for any circuit is given by:

$$Q = 2\pi \left(\frac{\text{Maximum Energy Stored}}{\text{Power Dissipated Per Cycle}} \right)$$

Question 3

Prove that energy stored in the capacitor equals energy stored in the inductor at resonance in a parallel RLC circuit.

Solution:

We know that energy stored in capacitor and inductor are given by:

$$E_L = \frac{1}{2}LI^2, \quad E_C = \frac{1}{2}CV^2$$

At resonance,

$$\omega = \omega_0 = \frac{1}{\sqrt{LC}}$$

Since $V_C = V_L = V$, we calculate I_L :

$$I_L = \frac{V}{X_L} = \frac{V}{\omega L} = V \frac{\sqrt{LC}}{L} = V \sqrt{\frac{C}{L}}$$

Now, the energy stored in the inductor:

$$E_L = \frac{1}{2}L \left(V^2 \frac{C}{L} \right) = \frac{1}{2}CV^2 = E_C$$

Thus, energy stored in capacitor equals energy stored in inductor.

$$E_L = E_C$$

Acknowledgments

We would like to express our gratitude to the following contributors for their efforts in preparing this document:

- **Evan:** Structuring and formatting of the $\text{\LaTeX}$ document.
- **Aditya Murugesb:** Contribution to Lecture 1 and 2.
- **Ayachi Mishra:** Contribution to Lecture 3.
- **Aditya Mohan:** Contribution to facts and trivia.